

Theoretical Analysis

Gear Calculations:

1. Determine the gear ratios for each stage of reduction.

Input speed = 1500 rpm

Output speed = 300 rpm

Total reduction ratio (R_t) = $1500 / 300 = 5$

Gear ratio at stage 1 (R_1) = 2.5

Gear ratio at stage 2 (R_2) = 2

$$R_t = R_1 \times R_2 = 2.5 \times 2 = 5$$

2. Calculate the number of teeth for each gear.

For stage 1,

gear 1 teeth = T_1

gear 2 teeth = T_2

$$R_1 = T_2 / T_1 = 2.5 \Rightarrow T_2 = 2.5 \times T_1$$

Let's take $T_1 = 20$,

$$T_2 = 2.5 \times 20 = 50$$

For stage 2,

gear 3 teeth = T_3

gear 4 teeth = T_4

$$R_2 = T_4 / T_3 = 2 \Rightarrow T_4 = 2 \times T_3$$

Let's take $T_3 = 20$,

$$T_4 = 2 \times 20 = 40$$

Shaft and Bearing Calculations:

1. Calculate the torque on each shaft.

Torque on shaft 1, T_1 = Motor output torque = 90 kg cm = 8.82 Nm

Torque on shaft 2, $T_2 = T_1 \times R_1 = 8.82 \times 2.5 = 22.05$ Nm

Torque on shaft 3, $T_3 = T_2 \times R_2 = 22.05 \times 2 = 44.1$ Nm

Housing and Material Analysis:

1. Justify the material selection for gears and housing based on mechanical properties, cost, and manufacturability.

Gears:

Material: AISI 1045 Steel, cold drawn

Properties: Good balance of strength, hardness, and cost-effectiveness.

Housing:

Material: Aluminum Alloy (6061-T6)

Properties: Lightweight, good thermal conductivity, easy to manufacture.